

Summary Data on Acid Cropland and Lime Requirements in Kenya

GAIA

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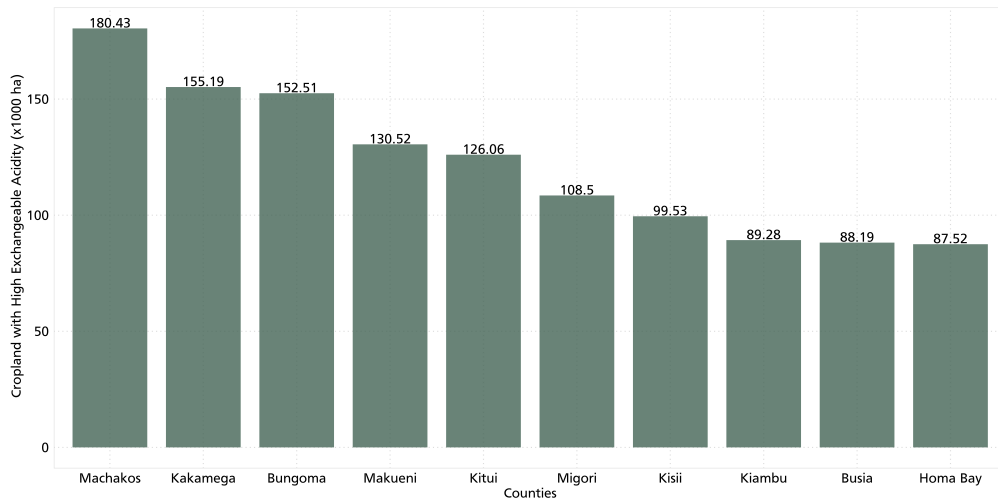
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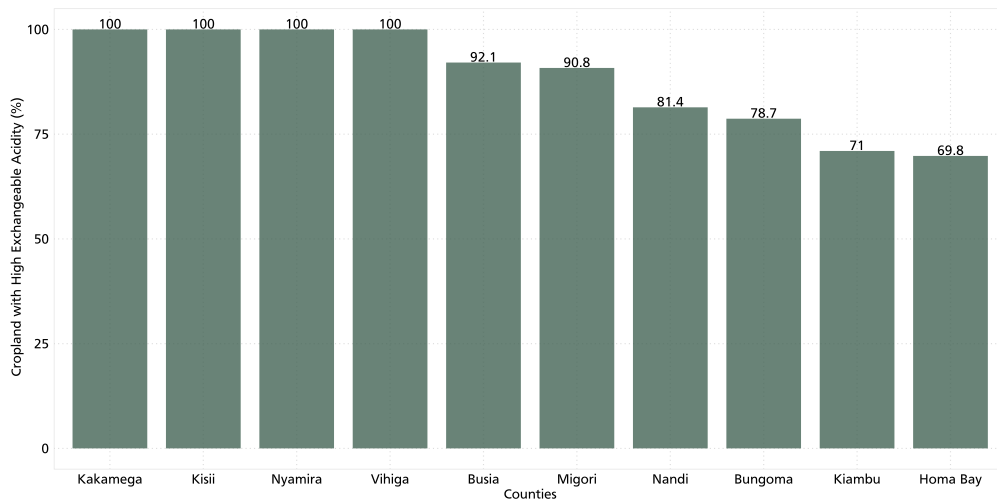
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Kenya**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production and production value from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

Top 10 Counties in Kenya with the highest acidic cropland



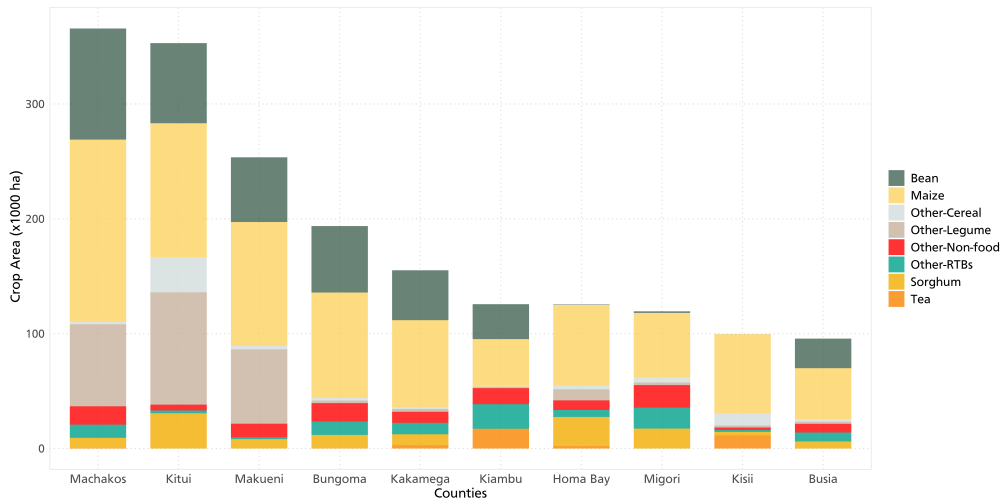
Top 10 Counties in Kenya with the highest acidic cropland (Percentage of total cropland)



└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Counties, not segregated by acidic cropland

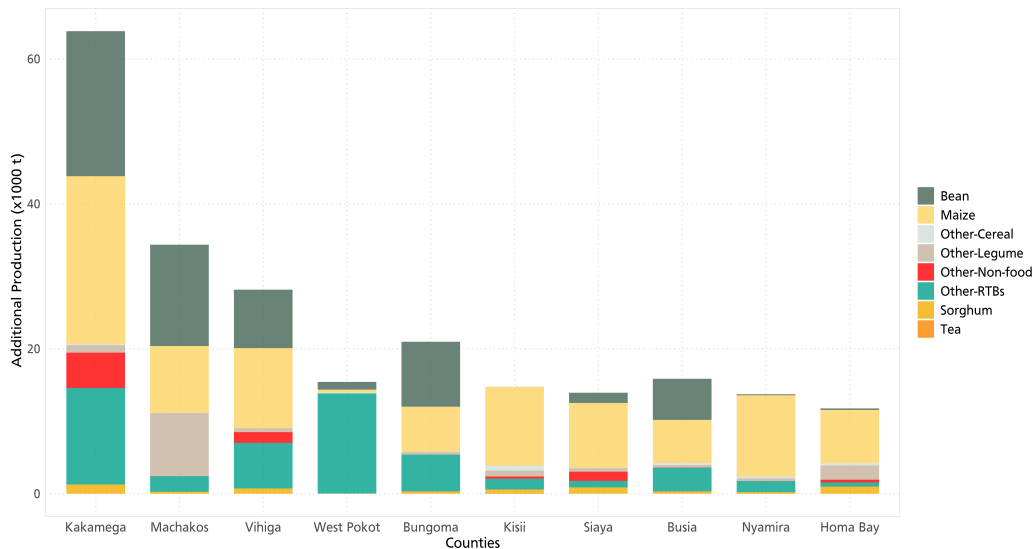
Total crop area in the top-10 most acidic Counties, not segregated by acidic cropland



Total crop area in the top-10 most acidic Counties, not segregated by acidic cropland (table)

Counties	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum	Tea
Bungoma	57.88	91.63	2.46	2.17	16.02	11.69	11.74	0.11
Busia	25.79	44.29	2.39	1.76	7.52	7.78	6.16	0.03
Homa Bay	0.46	70.27	3.15	9.63	8.31	6.11	25.39	2.11
Kakamega	43.44	75.78	1.50	2.48	9.64	9.96	9.53	2.86
Kiambu	30.46	41.28	0.47	0.92	13.97	21.37	0.09	17.14
Kisii	0.09	68.52	10.79	1.86	2.10	1.70	2.99	11.48
Kitui	69.77	116.53	30.68	97.76	5.08	2.48	30.68	-
Machakos	96.77	158.79	1.99	71.39	15.99	11.50	8.94	0.37
Makueni	56.32	107.91	2.99	64.68	11.84	1.63	8.16	-
Migori	1.41	56.27	4.37	2.25	19.58	18.24	17.37	-

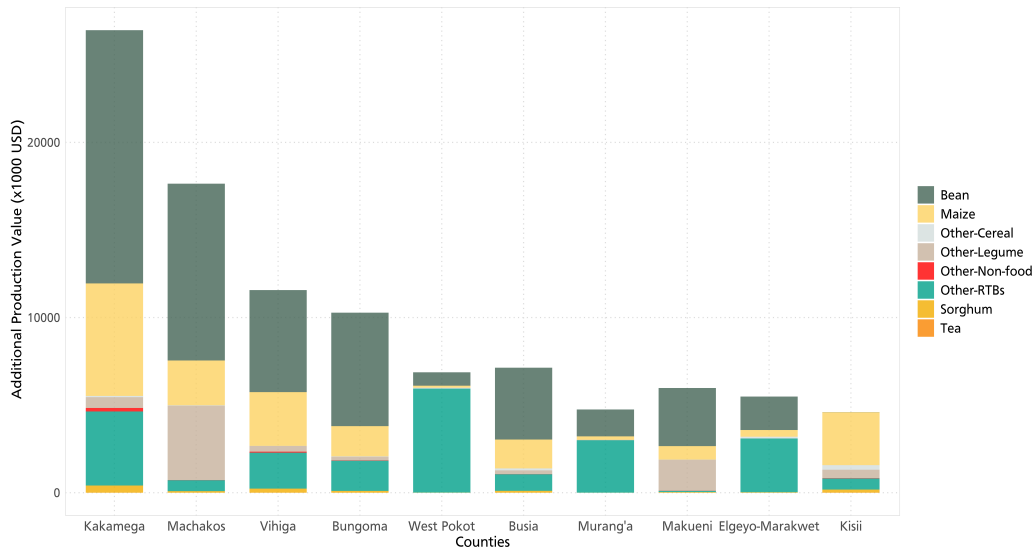
Potential for additional production (tone) in top 10 acidic Counties in Kenya



Potential for additional production (1000 tone) in top 10 acidic Counties in Kenya(Table)

Counties	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum	Tea
Bungoma	8.96	6.21	0.06	0.35	0.01	5.06	0.32	-
Busia	5.67	5.93	0.30	0.35	0.02	3.26	0.33	-
Homa Bay	0.21	7.35	0.32	1.95	0.38	0.57	0.97	0.01
Kakamega	20.01	23.14	0.18	1.04	4.86	13.32	1.26	0.01
Kisii	0.01	10.86	0.70	0.79	0.32	1.48	0.53	0.05
Machakos	13.98	9.20	0.06	8.69	0.02	2.14	0.27	-
Nyamira	0.14	11.18	0.29	0.37	-	1.52	0.16	0.05
Siaya	1.41	8.95	0.04	0.49	1.28	0.90	0.87	-
Vihiga	8.07	11.05	0.01	0.54	1.49	6.28	0.72	0.01
West Pokot	1.06	0.40	0.13	-	-	13.80	0.03	-

Potential for additional production value (US\$) in top 10 acidic Counties in Kenya

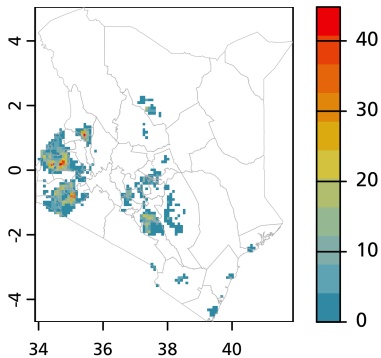


Potential for additional production value (1000 US\$) in top 10 acidic Counties in Kenya(Table)

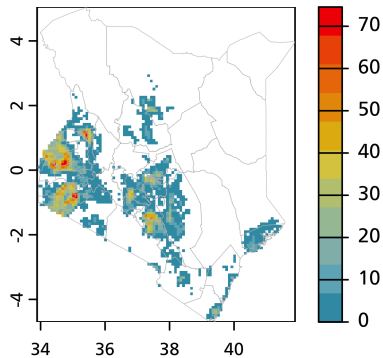
Counties	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum	Tea
Bungoma	6475.9	1724.5	23.2	207.6	19.2	1723.7	103.8	-
Busia	4098.9	1646.7	113.4	214.2	9.1	947.5	105.0	-
Elgeyo-Marakwet	1911.2	379.0	100.6	2.7	0.2	3070.4	24.8	-
Kakamega	14457.4	6427.1	70.4	616.1	199.3	4229.0	407.4	1.1
Kisii	10.3	3017.0	266.4	475.8	21.2	632.2	171.0	8.9
Machakos	10098.2	2556.4	21.8	4244.5	9.8	627.5	86.9	-
Makueni	3319.8	767.5	9.4	1770.8	7.6	70.6	33.6	-
Murang'a	1531.1	196.8	1.5	24.0	1.2	2993.8	0.8	-
Vihiga	5826.2	3068.5	4.7	325.6	60.3	2050.5	231.6	1.9
West Pokot	765.9	110.8	52.7	0.7	0.1	5933.8	9.6	-

Yield loss due to soil acidity for Maize and Bean in Kenya

Maize Yield Loss (%)

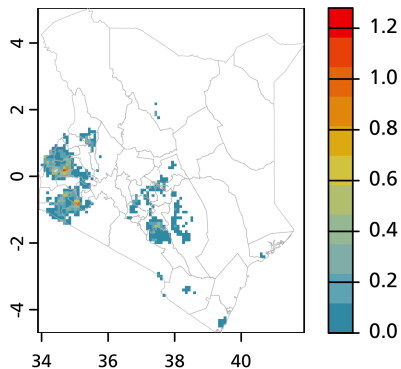


Bean Yield Loss (%)

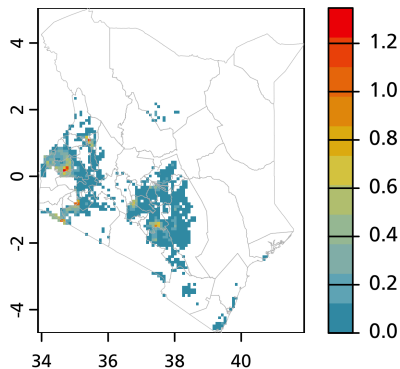


Yield response to lime application in Kenya

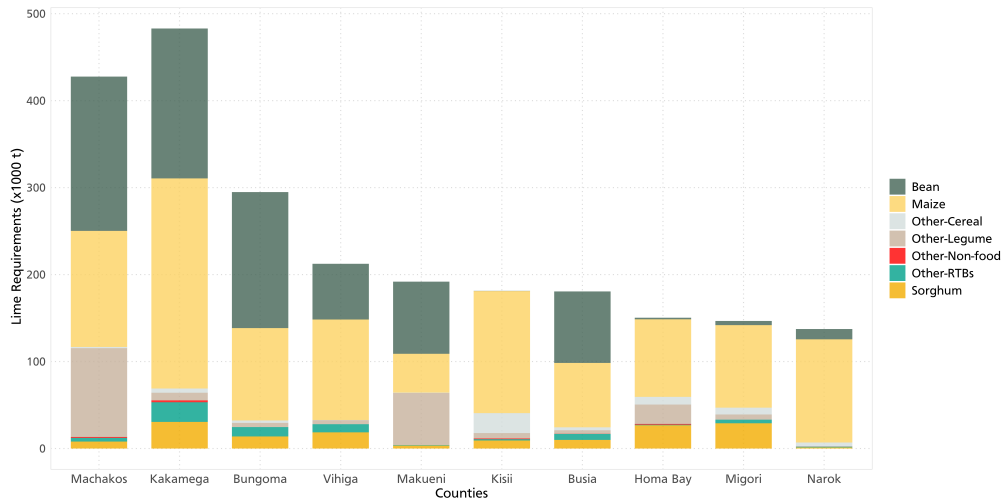
Maize Yield Response (t/ha)



Bean Yield Response (t/ha)



Lime requirements (t) for acid soil remediation in Kenya

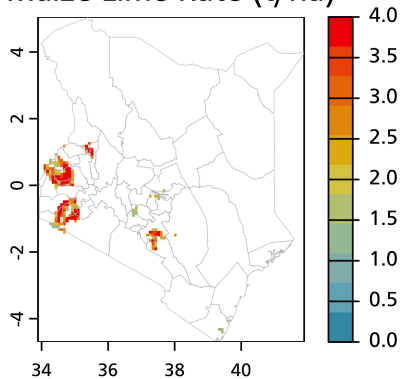


Lime requirements (1000 t) for acid soil remediation in Kenya (table)

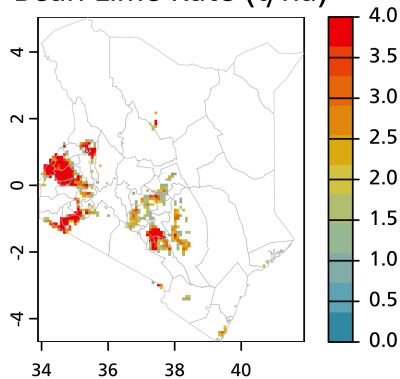
Counties	Bean	Maize	Other-Cereal	Other-Legume	Other-RTBs	Sorghum	Other-Non-food
Bungoma	156.36	106.23	2.80	4.87	10.77	13.84	-
Busia	82.24	73.98	3.36	4.12	6.88	9.94	0.09
Homa Bay	1.82	89.22	8.86	22.24	0.62	26.72	0.93
Kakamega	172.40	241.52	4.95	8.74	22.57	30.60	2.20
Kisii	0.26	140.27	22.97	6.20	1.48	9.23	0.85
Machakos	177.47	133.39	1.46	102.05	4.23	8.00	1.08
Makueni	82.97	44.61	0.48	59.93	0.58	3.34	-
Migori	4.82	94.85	7.88	5.96	4.19	29.00	-
Narok	11.86	118.57	4.42	0.44	0.86	1.17	0.06
Vihiga	64.09	115.41	0.31	4.62	9.35	18.60	-

Lime rate for Maize and Bean in Kenya

Maize Lime Rate (t/ha)



Bean Lime Rate (t/ha)



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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